SOFTWARE CHANGE
IMPLEMENTATION REPORT

CR-033: Last Login Tracking and Display for Student Accounts

Field	Details
Change Request ID	CR-033
Change Request Title	Last Login Tracking and Display for Student Accounts
Student Name	N. Bavanika
Student ID	JFICT24_19
Implementation Status	Completed
Date	13 October 2026

1. Change Request Summary

1.1 Business Requirement

Administrators can view student account and profile information, but the Students table did not previously display when a student last logged in. This change introduces a Last Login tracking feature to record and display the date and time of each student's most recent successful login.

The main objectives are:

- Record the date and time of each successful student login.
- Store the last login timestamp in the database.
- Retrieve last login information with student records.
- Display the last login date and time in the administrator Students table.
- Display “Never logged in” when a student has no recorded successful login.
- Ensure unsuccessful login attempts do not update the last login timestamp.
- Preserve existing authentication, authorization, and student management functionality.

1.2 Existing Behaviour

Before implementation:

- The login process validated the username and password and generated a JWT after successful authentication.
- The users table stored account and role information, but the last successful login time was not recorded for display.
- The administrator Students table displayed Registration No., Name, Email, Phone, and Actions.
- Administrators could not view the most recent successful login date and time from the Students table.
- Students without a previous successful login could not be easily identified through the dashboard.

1.3 Requested Change

Add a Last Login field to track and display the most recent successful login time for each student account.

The implementation must:

- Add a nullable last_login DATETIME column to the users table.
- Update last_login only after successful username and password verification.
- Store the current database date and time using NOW().
- Return last_login with student records through the existing student API.
- Display a Last Login column in the administrator Students table.
- Format the stored timestamp into a readable date and time.
- Display "Never logged in" when no login has been recorded.
- Preserve existing student Add, Edit, Delete, and Search functionality.
- Maintain the existing authentication and authorization flow.

2. Implementation

2.1 Implementation Summary

CR-033 was implemented across the database, backend authentication flow, student data retrieval, and administrator frontend.

A nullable last_login column was added to the users table. After the submitted password is successfully verified using bcrypt.compare(), the updateLastLogin() function updates the user's timestamp using the database NOW() function.

The student model retrieves the last login timestamp by joining the students table with the related users table. The existing student API returns this information to the frontend.

In DashboardPage.jsx, a new Last Login column displays the recorded timestamp in a readable format. When the value is NULL or no login has been recorded, the interface displays "Never logged in".

The existing login response, JWT authentication, authorization, and student CRUD functionality remain unchanged.

2.2 Affected Files

File	Change	Purpose
File / Component	Change	Purpose
database/schema.sql	Modified	Add the nullable last_login column to the users table.
backend/models/userModel.js	Modified	Add the updateLastLogin() function and retain existing user lookup functions.
backend/controllers/authController.js	Modified	Update last_login after successful password verification.
backend/models/studentModel.js	Modified	Retrieve users.last_login through a JOIN in student queries.
frontend/src/pages/DashboardPage.jsx	Modified	Add the Last Login column, format the timestamp, and display the fallback message.
Existing student API	Updated response	Include last_login through the existing GET /api/students/ endpoint.
Backend authentication and authorization	Preserved	Maintain existing password verification, JWT generation, and role-based access.
Database authentication data	Preserved	No password hashes or other sensitive

		authentication data are exposed through the student list.

2.3 Technical Changes

A nullable last_login DATETIME column was added to the users table.

The field stores the most recent successful login timestamp. The column allows NULL for accounts that have not completed a successful login.

The database timestamp is updated using:

UPDATE users

SET last_login = NOW()

WHERE user_id = ?;

The ? placeholder is used with a parameterized query in the backend application. It should not be executed directly in phpMyAdmin without replacing the placeholder with an actual user ID.

Password Verification and Timestamp Update

The existing authentication flow continues to verify the submitted password using bcrypt.compare().

After successful password verification, the controller calls:

await updateLastLogin(user.user_id);

The timestamp is not updated when the username is invalid or the password does not match.

Student Data Retrieval

The student model was modified to retrieve last_login from the related user record.

The existing GET /api/students/ endpoint returns student information along with the last login timestamp. No additional API endpoint was required.

Administrator Dashboard

The administrator Students table was updated to include a Last Login column.

The frontend:

- Displays the recorded login date and time.
- Shows “Never logged in” when last_login is NULL or unavailable.
- Preserves the existing student table structure and actions.
- Continues to use student_id for the Edit and Delete operations.

2.4 Implementation Logic

- The user opens the Login page.
- The user enters a username and password.
- The backend validates the submitted credentials.
- The system retrieves the matching user record.
- The submitted password is verified using bcrypt.compare().
- If the credentials are invalid, the login is rejected and last_login remains unchanged.
- If password verification succeeds, updateLastLogin() is called with the corresponding user_id.
- The database updates last_login using NOW().
- The existing JWT token is generated and returned through the existing login response.
- The student model retrieves the last login timestamp using the relationship between the students and users tables.
- The existing student API returns the timestamp with the student records.
- The administrator dashboard displays the Last Login column.
- The frontend formats the timestamp into a readable date and time.
- If no timestamp exists, "Never logged in" is displayed.
- Existing authentication, authorization, Add, Edit, Delete, and Search functionality continues to work.

2.5 Error and Data Handling

Existing error handling and authentication behaviour were preserved.

For Login:

- Empty username and password validation remains unchanged.
- Invalid username or password continues to return the existing authentication error.
- The timestamp is updated only after successful password verification.
- The existing JWT generation and login response remain unchanged.
- The existing Administrator and Student role-based navigation remains functional.

For Student Management:

- The student API retrieves last_login from the related user record.
- Students without a recorded login display "Never logged in".
- Existing student records retain their original student_id.
- Existing Add, Edit, Delete, and Search operations remain unchanged.
- Password hashes and other sensitive authentication data are not included in the student list response.

No login history table or additional authentication mechanism was introduced.

3. Testing & Verification

3.1 Testing Approach

- Testing Environment: Development
- Browser: Google Chrome

- Test Role: Administrator and Student
- Test Date: 13 October 2026

The following testing activities were performed:

- Functional testing
- Positive and negative login testing
- Database timestamp verification
- Student API response testing
- Frontend date and time display testing
- NULL value and “Never logged in” testing
- Authentication and authorization regression testing
- Student Add, Edit, Delete, and Search regression testing
- UI testing
- Error handling testing

3.2 Test Cases

Test ID	Test Case	Expected Result	Actual Result	Status
TC-01	Successful student login	last_login is updated with the current date and time.	Timestamp was updated after successful login.	Pass
TC-02	Incorrect password	Login is rejected and last_login remains unchanged.	Login was rejected and the timestamp remained unchanged.	Pass
TC-03	Invalid username	Login is rejected without updating the timestamp.	Login was rejected and the timestamp was not updated.	Pass
TC-04	Load student list	Student records include last_login.	Last login information was returned with student data.	Pass
TC-05	Display Last Login column	The administrator Students table displays the new column.	Last Login column was displayed correctly.	Pass
TC-06	Display readable timestamp	Date and time are formatted for administrator viewing.	Timestamp was displayed in a readable format.	Pass
TC-07	Student with no previous login	Display “Never logged in”.	The fallback message was displayed correctly.	Pass
TC-08	Multiple successful logins	Timestamp reflects the most recent successful login.	The latest successful	Pass

			login time was displayed.	
TC-09	Existing Add Student operation	A new student can be created successfully.	Student creation worked correctly.	Pass
TC-10	Existing Edit action	The correct student can be edited using student_id.	Edit functionality worked correctly.	Pass
TC-11	Existing Delete action	The correct student can be deleted using student_id.	Delete functionality worked correctly.	Pass
TC-12	Existing Search function	Student search continues to work.	Search functionality worked correctly.	Pass
TC-13	Valid login and JWT response	Existing authentication and token generation continue to work.	Login and token generation worked correctly.	Pass
TC-14	Administrator authorization	Existing administrator access rules remain unchanged.	Authorization behaviour was preserved.	Pass
TC-15	Student login and dashboard	Student login and role-based dashboard navigation continue to work.	Student login and dashboard navigation worked correctly.	Pass

3.3 Test Summary

Test Result	**Count**
Total Test Cases	15
Passed	15
Failed	0
Pending / Not verified	0
Pass Rate	100% Pass

4. Acceptance Criteria Verification

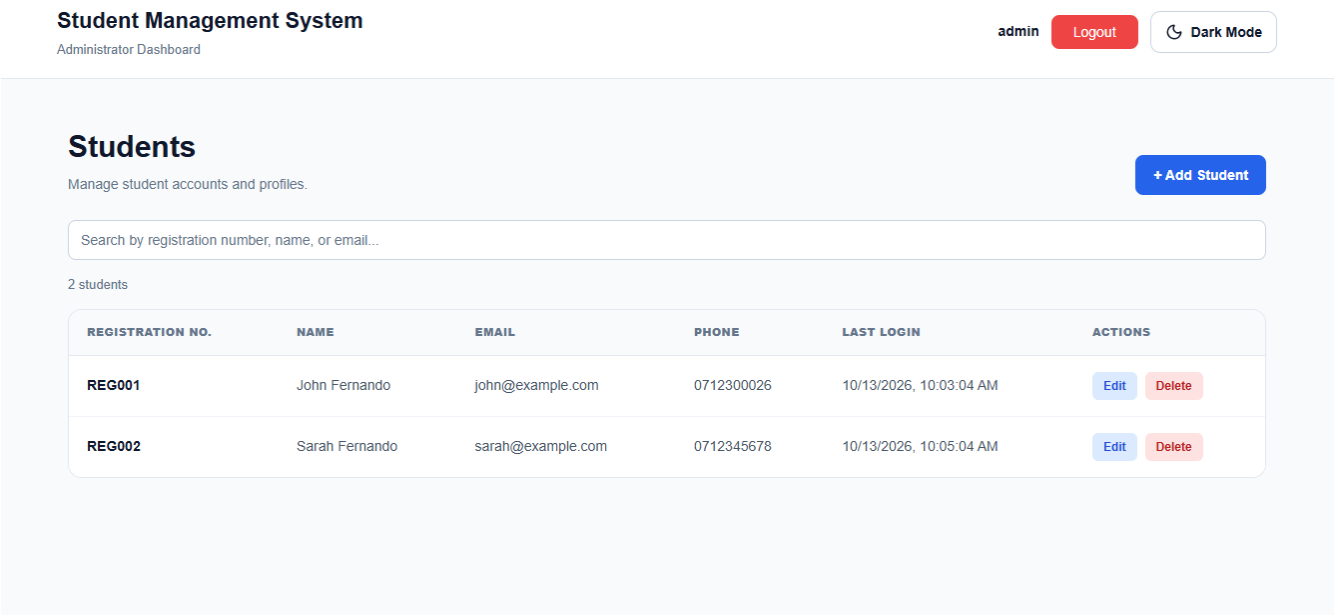
ID	Acceptance Criterion	Verification	Status
AC-001	Successful login records the latest login date and time.	Successful login and database verification.	Pass
AC-002	Incorrect login credentials do not update last_login.	Invalid password testing.	Pass
AC-003	The administrator Students table displays a Last Login column.	Dashboard UI verification.	Pass
AC-004	Students without a recorded login display “Never logged in”.	NULL timestamp display test.	Pass
AC-005	The existing student API returns last_login.	GET /api/students/ response verification.	Pass
AC-006	Existing Add, Edit, and Delete operations continue to work.	CRUD regression testing.	Pass
AC-007	Existing authentication and authorization remain unchanged.	Login and role-based access testing.	Pass
AC-008	Password hashes and sensitive authentication data are not exposed.	API response inspection.	Pass

5. Evidence

Evidence 01 – Administrator Students Table

Figure 1: Administrator Students table displaying the Last Login column with readable date and time values.

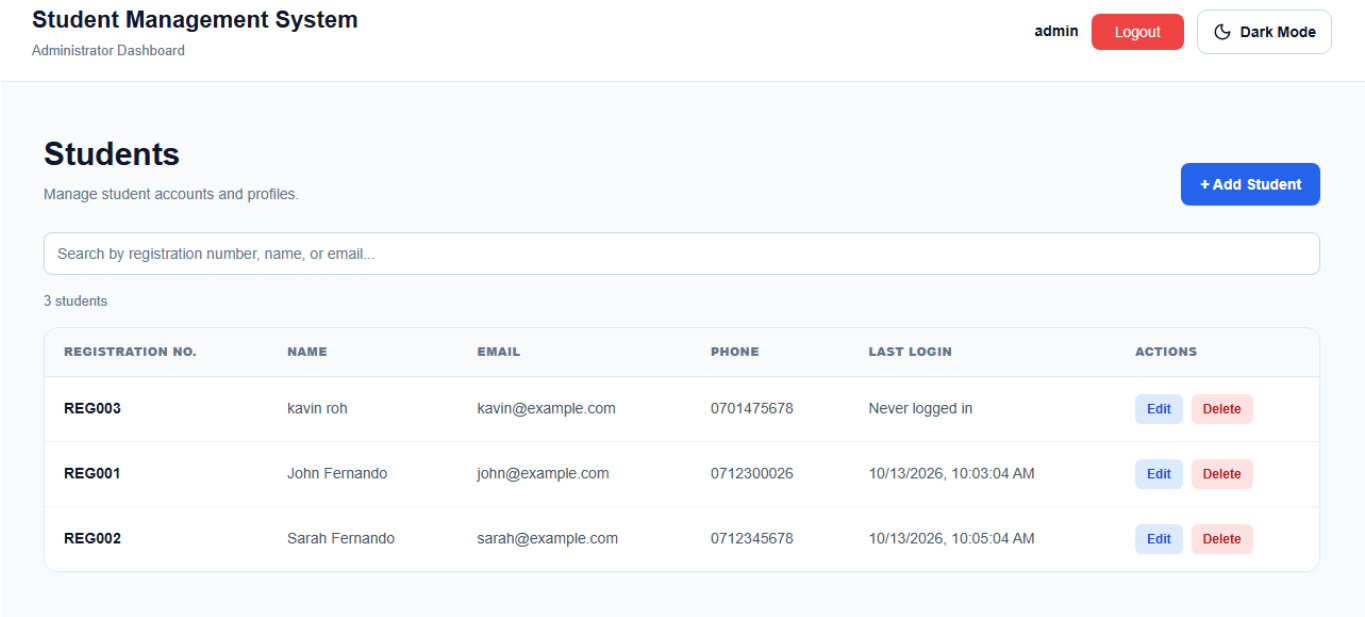
Related acceptance criteria: AC-003, AC-004, AC-005.



Evidence 02 – Successful Student Login

Figure 3: Student account with a NULL last_login value displaying “Never logged in” in the administrator Students table.

Related acceptance criterion: AC-004.



6. Final Result

Item	Result
Requirements	Completed
Database Change	Completed
Backend Login Timestamp Update	Completed
Student API Update	Completed
Administrator Last Login Column	Completed
Timestamp Formatting	Completed
“Never logged in” Fallback	Completed
Testing	15 of 15 Passed
Acceptance Criteria	All Met
Existing Authentication	Verified
Existing Authorization	Verified
Existing Student CRUD Functionality	Verified
New API Endpoint	Not Required
Final Status	Completed

6.1 Final Summary

CR-033 was successfully implemented to record and display the most recent successful login time for each student account in the Student Management System.

A nullable last_login column was added to the users table to store the timestamp. The backend updates this field only after successful password verification using the existing bcrypt authentication process. The timestamp is recorded using the database NOW() function.

The existing student API was extended to return the last login timestamp through the relationship between the students and users tables. The administrator dashboard now displays a Last Login column containing a readable date and time. For student accounts without a recorded successful login, the interface displays "Never logged in".

The existing JWT authentication flow, role-based authorization, and student Add, Edit, Delete, and Search functionality were preserved. No new API endpoint or login history table was introduced.

A total of 16 test cases were completed, with all 16 passing and no failures. All defined acceptance criteria were verified and marked as met.

The implementation improves account activity visibility for administrators while maintaining the existing functionality and security of the Student Management System.

6.2 Issues / Limitations

- The system records only the most recent successful login and does not maintain a history of previous logins.
- Accounts without a successful login retain a NULL last_login value and display "Never logged in".
- The database schema change must be applied in each environment before deploying the updated application.
- The displayed timestamp depends on the database/application timezone and the frontend date formatting.
- No outstanding issues were identified for the requested Last Login feature.

Student Name: N. Bavanika

Student ID: JFICT24_19

Date: 13 October 2026

Signature:

~ END OF REPORT ~